HDB Resale Solutioning of ETL pipeline
(January 2012 to December 2016)

Version 1.0

Action	Name, Designation	Date
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Reviewed by:		

Version	Effective Date	Summary of Changes	Author
0.1	26 Aug 2025	Initial Draft	Joe Chinnappan

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1 Requirement

The requirement, the **Data Engineering Team** is responsible for building a repeatable Python-based ETL pipeline that prepares HDB resale flat transactions (January 2012 to December 2016) for the **Data Science Team** to perform price trend analysis and forecasting.

2 Objective

Design and implement a Python-based ETL pipeline to:

1. Extract HDB resale flat datasets (Jan 2012 to Dec 2016).
2. Clean and standardize the data.
3. Transform the data into an analytics-ready format.
4. Store the curated dataset for downstream Data Science consumption.
5. Support automated and repeatable execution.

3 Dataset Extraction

Following are the source data from HDB in csv format. Total **92,544** are scope of Extraction, after extraction **92,544** are stored in **combined.csv** file

File Name	Scope of Extraction	Number of Records
ResaleFlatPricesBasedonApprovalDate19901999.csv	N	N/A
ResaleFlatPricesBasedonApprovalDate2000Feb2012.csv	From Jan 2012	3,188
ResaleFlatPricesBasedonRegistrationDateFromMar2012toDec2014.csv	Y	52,203
ResaleFlatPricesBasedonRegistrationDateFromJan2015toDec2016.csv	Y	37,153
ResaleflatpricesbasedonregistrationdatefromJan2017onwards.csv	N	N/A

4 Dataset Cleaning

As part of the Data cleansing, following activities were performed

1) Missing Values,
2) Duplicates &
3) Formatting & Data Validation.

Validation Rules

Field Name	Validation Rule	Invalid Records	Observation
month	Valid Date in Format YYYY-MM	0	
town	Alphabetic Characters	2744	Failed for KALLANG/WHAMPOA Note: As of now ignored as it is valid Town Can be handled by "^[A-Z\s\-'/.&]+$"
flat_type	Alphanumeric Characters	0	
flat_model	Alphanumeric Characters	0	
storey_range	Format NN TO NN	0	

5 Dataset Transformation

Following Data Transformation were performed

1) Recomputed the based on the **Lease_Period in terms of Years and Months** (**99 years – (today- lease_commence_date)**, assume that HDB lease is 99 years
2) If any **duplicate** based on **all columns have the exact same value**, except for the resale price take the higher price and discard other records
3) Additional column for **Resale Identifier** based on the section **"Data Transformation Requirements"**.
4) Additional column for **Hash value** this identifier column using an irreversible hashing algorithm based on **lambda hashlib.sha256**

6 Dataset Loading

After Dataset Transformation is completed, all the required fields loaded into the final Dataset**, hdb_with_resale_identifier_hashed.csv** for Data analysts' team.

7 Data Profiling

For data profiling used **Sweetviz** library**. Sweetviz** is an open-source python library that generates beautiful, high-density visualizations to kickstart EDA with a single line of Code. Sweetviz is taken as an inspiration from Pandas-Profiling. Data Profiling report is stored in **Data_Profiling_Report_HDB.html**.

The 2 main functions of the reports are

1. Analyze the data
2. Compare the data

8 Post Verification

8.1 Record Count Reconciliation

Description	Record Count	Output File Name
Original Rows From Jan 2012 to Dec 2016	92,544	Combined.csv
Duplicates Rows	273	duplicate_records.csv
Original Rows, after removing Duplicate	92,271	cleaned.csv

Test Data for Field Level validation

month	town	flat_type	block	storey_range	floor_area_sqm	lease_commence_date	resale_price	remaining_lease	avg_resale_price	Resale_Identifier
2012-01	ANG MO KIO	2 ROOM	406	01 TO 03	44	1979	257800	51 years 6 months	256966.67	S4062501A
2012-01	ANG MO KIO	2 ROOM	314	07 TO 09	44	1978	263000	50 years 6 months	256966.67	S3142501A
2012-01	ANG MO KIO	2 ROOM	314	10 TO 12	44	1978	275000	50 years 6 months	256966.67	S3142501A
2012-01	ANG MO KIO	2 ROOM	170	01 TO 03	45	1986	260000	58 years 6 months	256966.67	S1702501A
2012-01	ANG MO KIO	2 ROOM	174	07 TO 09	45	1986	226000	58 years 6 months	256966.67	S1742501A
2012-01	ANG MO KIO	2 ROOM	508	04 TO 06	44	1980	260000	52 years 6 months	256966.67	S5082501A
2012-01	ANG MO KIO	4 ROOM	309A	01 TO 03	90	2006	528000	78 years 6 months	475595.13	S3094701A
2012-01	BUKIT BATOK	4 ROOM	289F	01 TO 03	92	1998	425000	70 years 6 months	438467.79	S2894301B
2012-01	BUKIT MERAH	1 ROOM	7	07 TO 09	31	1975	250000	47 years 6 months	245000	S0072401B
2012-01	BUKIT MERAH	2 ROOM	50	04 TO 06	42	1973	274000	45 years 6 months	358000	S0503501B

8.2 Field Level validation

1) **Field: month**

   Month Data, 2012-01 follows format **YYYY-MM** format.

2) **Field: Storey-range**

   Storey-range Data, 01 TO 03, follows format **NN TO NN** format.

3) **Field: Remaining lease**

Lease-commence date Is 1979,

Remaining-lease is 99 years-(2026 July – 1979 Jan) = **51 years 6 months**

4) **Field : avg_resale_price**

Based on the 3 fields month(**2012-01),**town(**ANG MO KIO**) & flat_type(**2 ROOM**).

avg_resale_price=SUM(resale_price)/Number of Resale House

(257800+263000+275000+260000+226000+260000)/6 = **256,966.67**

5) **Field :** Resale_Identifier

First character is “S is **S**4062501A

Next 3 digits is Block number 406 is S**406**2501A

Next 2 digits of avg_resale_price is = **256,966.67 is S**406**25**01A

Next 2 digits is Month(01) is S406**2501**A

Last Digit 1st character of tow n(ANG MO KIO) is S4062501**A**

6) **Field : Block number in Resale Identifier**

Additional verification if Block has Character and Block Number less than 3 digits.

Block	Resale_Identifier
309A	S**309**4701A
289F	S**289**4301B
7	S**007**2401B
50	S**050**3501B

9 Recommendation for additional validation

i) Drop records with missing `resale_price`

ii) Validate Resale Price Range like **resale_price < 50000 & resale_price > 2000000**

iii) Validate Remaining Lease **remaining_lease_years <= 0 or remaining_lease_years > 99**

iv) Flat Type Validation like **valid_types in ('1 ROOM', '2 ROOM', '3 ROOM','4 ROOM', '5 ROOM','EXECUTIVE', 'MULTI-GENERATION')**

10 Pre request before execution

i) Create Folder **C:\HDB_Project**(Root Folder)

ii) Create Folder “datasets” under **C:\HDB_Project(To store all 5 .csv HDB Dataset files)**

iii) Create Folder “output under” **C:\HDB_Project\datasets (to store output data files)**

iv) Copy the all the input .csv files in **C:\HDB_Project\datasets**

v) Run the Python script **HDBProjectResale.py**

vi) Following output files will create in **C:\HDB_Project\datasets\output**.

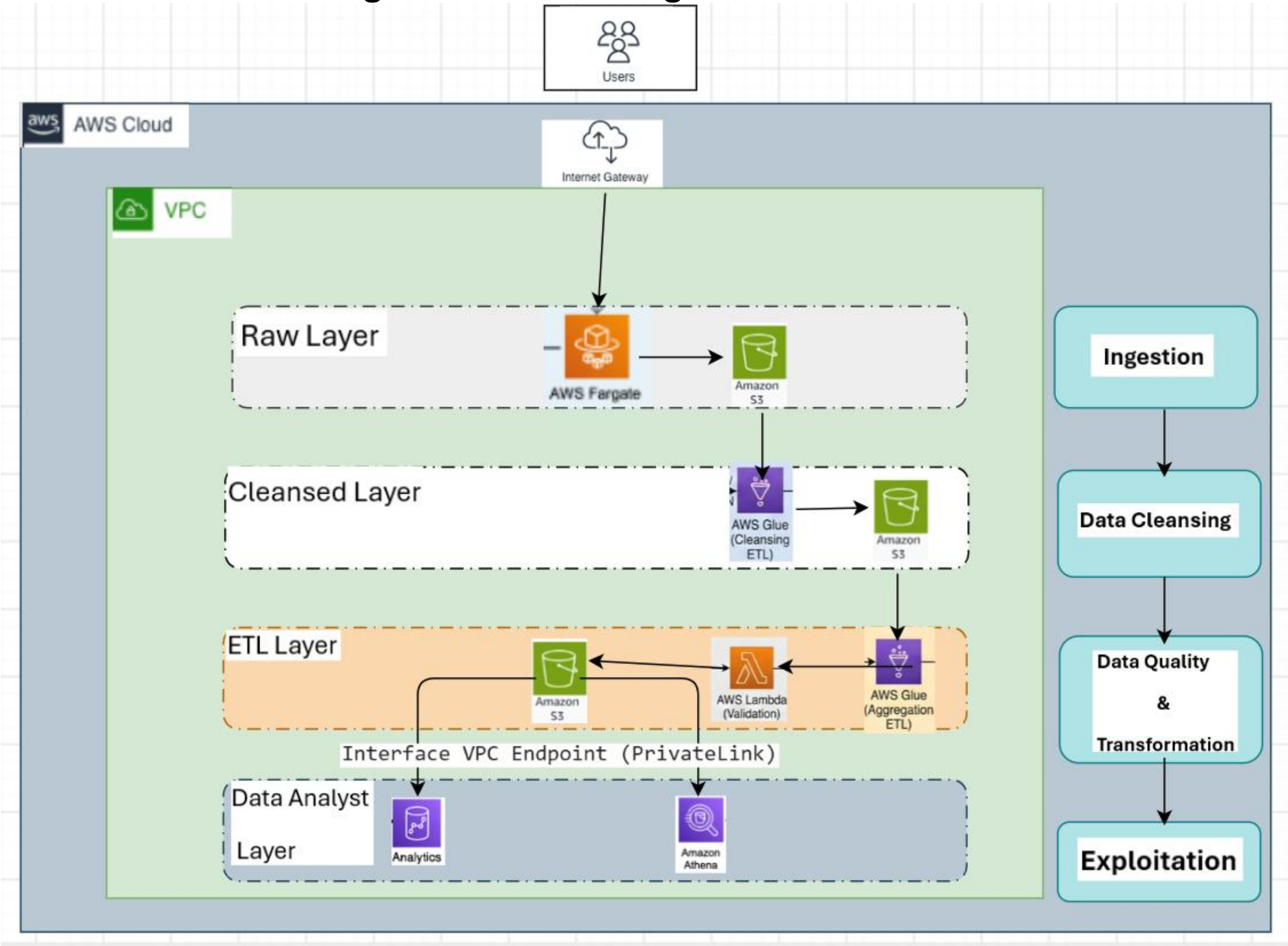
Sequence of the File Creation	**File Name(.csv)**	**Objective of the files**
1	combined	**Combining** all the records from HDB .csv files for the month **from '2012-01'(Jan 2012) to '2016-12'(Dec 2016)**
2	duplicate_records	**Duplicate** Records **from '2012-01'(Jan 2012) to '2016-12'(Dec 2016)**
3	Cleaned	**Clean Records** after **removing duplicate** Records from '2012-01'(Jan 2012) to '2016-12'(Dec 2016)
4	validation_report	Number of records **violates Validation Rules**
5	invalid_records	Contains invalid records which **violates Validation Rules**
6	cleaned_hdb_data	Clean data **with additional recalculating remaining_lease** in terms of Year and Month
7	hdb_with_resale_identifier	Clean data **with additional recalculating resale_identifier**
8	hdb_with_resale_identifier_hashed	Clean data **with additional recalculating resale_identifier hashed value**
9	Data_Profiling_Report_HDB.html	Contains the Data Profiled details

11 Global Variables

Following are the Global variables used in the Python Program

Variable Name	Variable Value	Description
folder	C:\\HDB_Project\\datasets	To keep all the HDB .csv Files
output_folder	C:\\HDB_Project\\datasets\\output	To store the output files
start_date	2012-01	To store the start Year and Month of the Extraction
end_date	2016-12	To store the end Year and Month of the Extraction

12 Architectural Design and Solutioning



12.1 Data Ingestion Requirements

1) **Batch data** ingestion from data.gov.sg. this design support large files transmission (e.g. > 100 MB)

Fargate provides:

- Elastic scaling
- Better support for large file transfers

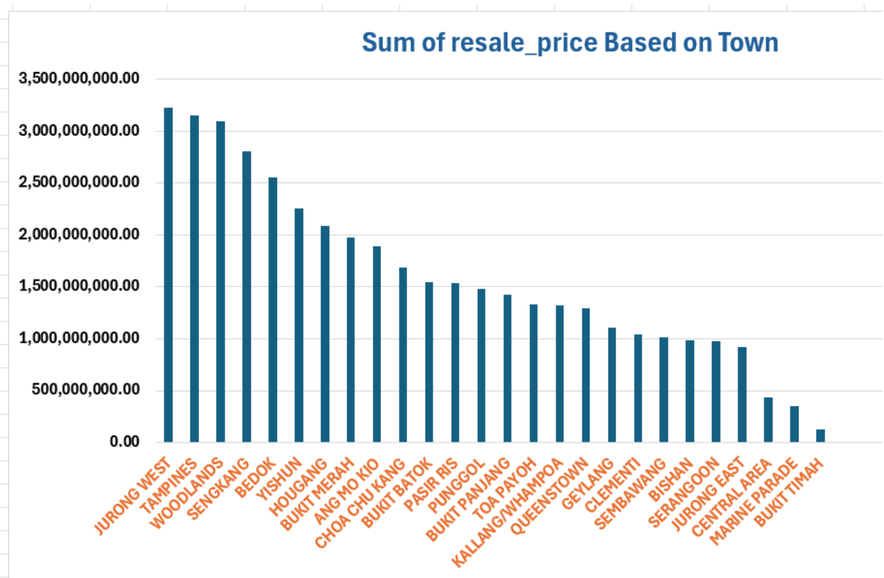
2) This **VPC acts** as a controlled buffer between the public internet and HDB's internal platform.

12.2 Data Exploitation Requirements

Query output reaches securely to Athena through Interface VPC Endpoint (Private Link).

13 Potentially Anomalous Resale Prices

i) Analysis based on the Sum of Sales against Towns



Based on the analysis, top 3 Resale Price are

1) Jurong West
2) Tampines
3) Woodlands

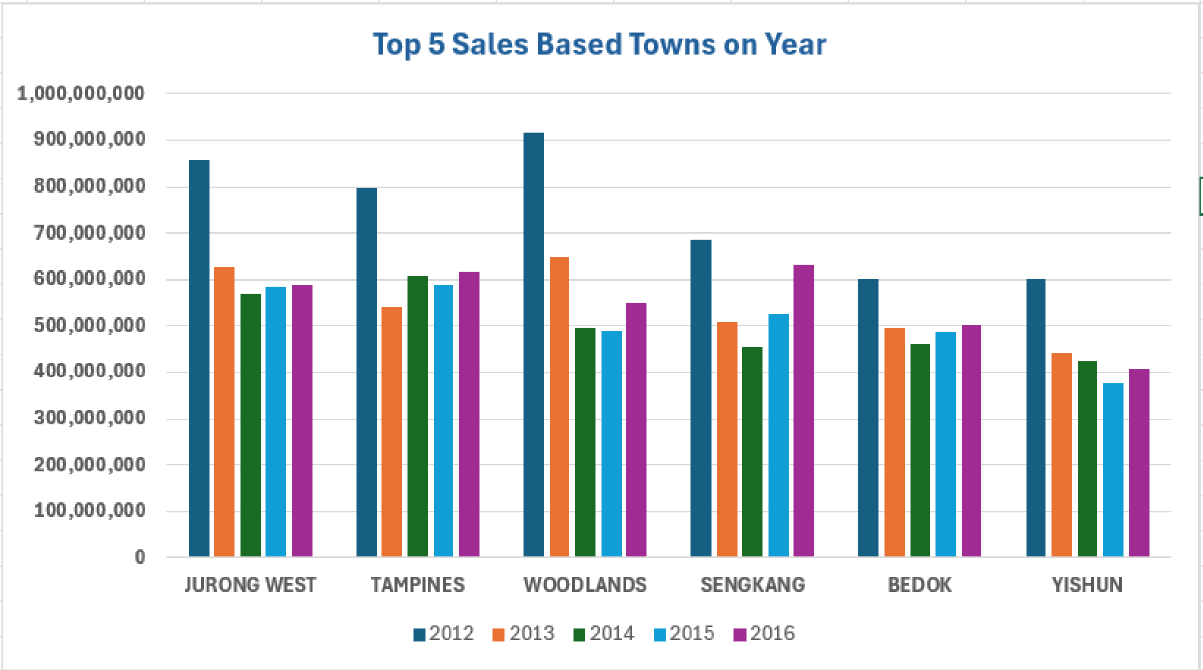
Based on the analysis, Bottom 5 Resale Price are

1) Bukit Timah

2) Marine Parade
3) Central Area

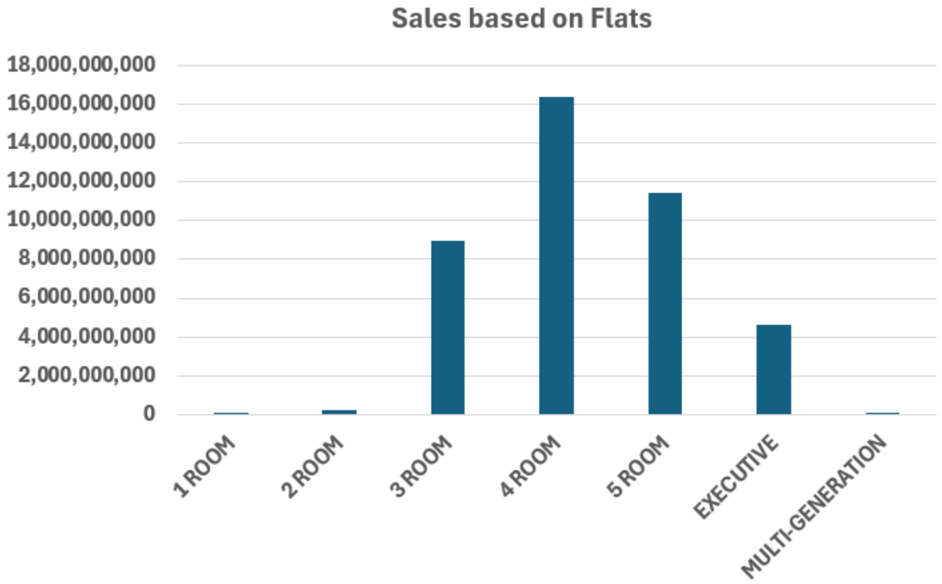
Conclusion: Most of the people are selling the house in outer area of Singapore except Bukit Timah not in the central region.

ii) **Analysis based on the Sum of Sales against each year for Top 5 Towns.**



Conclusion: Comparing the sales from 2012 to 2016. Sales is more in all the Towns in 2012. Remaining years there is no major sales fluctuation

iii) **Analysis based on the Sum of Sales Price based on Flats.**



Conclusion: Based on the Flats, maximum sales prices are in the order 1) **4 Rooms 2) 3 Rooms and 3) 5 Rooms**